Activity 2, Intermediate - Colby's Mouse Town

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In this adventure, students will learn introductory programming concepts by planning and navigating routes through Colby's town on graph paper. They will practice sequencing by breaking bigger tasks into smaller steps.

Setup - Activity 2

This activity can be broken down into several elements for lesson planning.

Objectives

- Understand that programs execute commands in sequence
- Break problems down into sequential steps
- Represent sequences and loops tactually

Materials Needed

- Graph paper with 1" grid
- Assorted Texture Paper
- Stickers
- Braille labels
- Printable block coding commands
- Scissors
- Glue stick
- Other art supplies
- Lego blocks
- Wheatley Tactile Diagramming Kit

Ideas for Carrying Out This Activity

- Encourage students to work with partners if possible

Adventure Map: Activity 2

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

- Ask your student: How can we break bigger tasks down into smaller steps? How can breaking down a larger problem into smaller problems make it easier to come up with a solution?
  - Review vocabulary with your student:
  - Algorithm - A list of steps that you can follow to finish a task
  - Sequence - In a program each step is an instruction to be performed. Sequence is the order in which the steps are carried out.
  - Loop - A sequence of instructions that is repeated a certain number of times.
- Show examples of simple programs drawn on graph paper.

2. Introduce Colby's Mouse Town

- For the introduction, reflect on the last adventure where students programmed Colby to navigate different paths to get to the cheese.
- For this exercise, we will use sheets of graph paper with a 1" grid to plan a layout for Colby's town. Starting at the upper left-hand corner, make this spot Colby's house. This will give you a starting spot for the next section. Have the student come up with other locations in the town, use places from the last activity, some places might be:
  - School
  - Library
  - Park
  - Bank
  - Cheese Shop
- Have the student indicate where each place is located on the grid.
  - They can use markers, stickers, textured paper, braille labels or any other art supplies you have.

3. Navigating the Town

- Go over the possible command you can give to Colby, see the printable block coding commands.
- Give your student a problem: Colby needs to go to school, what is the quickest route he can take?
  - Start out with an easy challenge and give your student harder and harder challenges.

  - Have some roadblocks or other obstacles you can add to the grid to change Colby's path.
  - After a couple routes, introduce the repeat command.

4. Sharing and Discussion

- What did we learn today?
- What if we have a bunch of the same commands in a row, what can we do to simplify our list of commands?

5. Conclusion

- Summarize the key points of the lesson.
- If time allows, have the students create another town or expand on the one they have. They could also create a grid for the inside of Colby's house or the inside of the school.
- You can also ask that students incorporate one or more loops in their algorithm.

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